

QC II - Part 2: Application

Lucas Hirschfeld, s6luhirs@uni-bonn.de

Supervisor: Benedikt Bädorf, Robin Dahl

First Submission: July 21, 2025

Contents

1	Electron Correlation	2
1.1	Multireference Methods	2
1.2	Carbenes	4
2	Basis Set Convergence	6
3	Thermochemistry	9
4	Kinetics	11
5	Solvation	14
6	Activation Energies	17
7	Noncovalent Interactions	19
8	Spectroscopy	21
8.1	IR-Spectroscopy	21
8.2	The Color of Indigo	23
8.3	NMR Parameters	24
9	Appendix	29
	References	30

1 Electron Correlation

1.1 Multireference Methods

The dissociation of Hartree-Fock (HF) was studied using different methods for H-F bond distances between 0.5 Å and 4.0 Å using the def2-TVZP basis set. The calculation was done using ORCA 5.0.4.^[1,2] The following methods were used: restricted Hartree-Fock (RHF), unrestricted Hartree-Fock (UHF), Møller-Plesset perturbation theory of second order (MP2), coupled cluster with single, double and perturbative triple excitations (CCSD(T)), CASSCF(2,2) and CASSCF(6,6). Complete active space self-consistent field (CASSCF)^[3] is a multireference method that was used in two different ways, first with two active electrons in two active orbitals (CASSCF(2,2)) and second with six active electrons in six active orbitals (CASSCF(6,6)).

Figure 1 shows the dissociation curves for the different methods. The calculated energies are calculated relative to the dissociation limit of the H-F molecule. The dissociation limit is defined as the energy of a hydrogen atom and a fluorine atom at infinite distance.

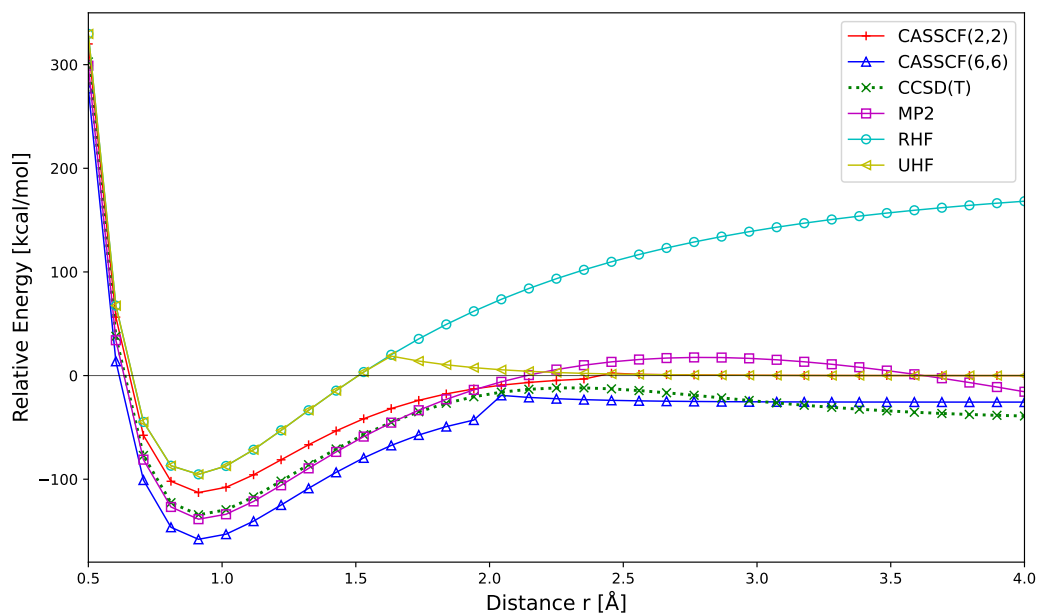


Figure 1: H-F Dissociation for various methods.

The RHF and UHF methods show a dissociation curve that is not suitable for describing the dissociation of H-F. The RHF curve approaches relative energies larger than 0 kcal/mol,

which is not physically meaningful. This is due to the problem that each pair of electrons needs to occupy the same spatial wave function. As a result of this, the ionic dissociation of H-F is not described correctly.

This problem is absent in the UHF curve, showing the correct dissociation limit. To obtain the best possible of the UHF curve, the calculation was done from 0.5 Å to 4.0 Å and from 4.0 Å to 0.5 Å and the lowest energy was taken for each distance. Nevertheless, the UHF curve shows a unphysical behaviour at 1.6 Å, where the relative energy has a maximum. This is due to the fact that the UHF method does not account for the electron correlation in the dissociation limit, which is caused by the way of how the calculation is done. Furthermore, both RHF and UHF methods calculate a binding energy at the equilibrium that is too low due to missing electron correlation.

Post HF methods like MP2 and CCSD(T) show a much better description of the dissociation of H-F, because they take the correlation effects into account. The MP2 energy correction is calculated as follows:

$$E_0^{(2)} = \frac{1}{4} \sum_{i,j,ab} \frac{|\langle ij||ab \rangle|^2}{\epsilon_i + \epsilon_j - \epsilon_a - \epsilon_b}, \quad (1)$$

where $\langle ij||ab \rangle$ are the two-electron integrals and ϵ_i , ϵ_j , ϵ_a and ϵ_b are the orbital energies of the occupied and virtual orbitals. However, this results that the MP2 energy correction approaches $-\infty$ for large distances, which is not physically meaningful. This occurs because the energy difference between occupied and virtual orbitals decreases for large distances, which leads to denominator approaching zero. A similar argument can be made to a lesser extent regarding CCSD(T), given the perturbative calculation of triple excitations.

Both CASSCF methods describe the dissociation of H-F correctly, but have an unphysical behavior at 2.5 Å for the CASSCF(2,2) and at 2.0 Å for the CASSCF(6,6) method, which is not as drastic as in the UHF curve. The CASSCF(2,2) method computed to high energies at the equilibrium distance, but the description of the dissociation limit is accurate with only 2 active electrons in 2 active orbitals.

1.2 Carbenes

The calculation of the singlet-triplet energy splitting ΔE_{S-T} for two carbenes, namely Methylene and *p*-Benzyne, and the Methylene H-C-H angles α_{HCH} were computed using the def2-TZVP basis set and the following methods: HF, MP2, CCSD(T), Becke 3-parameter exchange functional with Lee-Yang-Parr correlation functional (B3LYP), Tao-Perdew-Staroverov-Scuseria (TPSS) and Perdew-Wang 6-parameter exchange functional with Becke-95 correlation (PW6B95). The geometries for the CCSD(T) calculations were optimized using the MP2 method in a def2-TZVP basis set. The structure of the carbenes is shown in Figure 2.

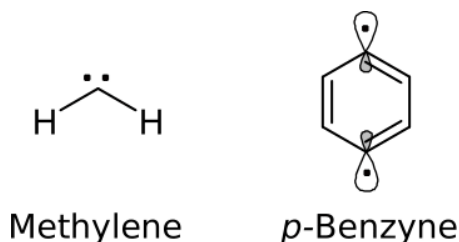


Figure 2: Structure of Methylene (left) and *p*-benzyne (right). Figure taken from [4].

The calculations for the geometry optimizations and the single point calculations were done using TURBOMOLE.^[5] The results for the singlet-triplet energy splitting ΔE_{S-T} and the Methylene H-C-H angles α_{HCH} are shown in Table 1.

Table 1: Singlet-Triplet energy splittings (kcal/mol) for *p*-benzyne and methylene along methylene H-C-H angles. All values were obtained using a def2-TZVP basis. CCSD(T) calculations were performed using MP2 geometries.

Method	ΔE_{S-T} [kcal/mol]		α_{HCH}	
	<i>p</i> -Benzyne	Methylene	Singlet	Triplet
HF	80.4	28.4	103.7°	131.8°
MP2	-28.6	15.8	102.4°	133.1°
CCSD(T)	-3.3	10.8	-	-
B3LYP	13.4	11.9	101.9°	135.1°
TPSS	7.0	17.8	101.3°	135.4°
PW6B95	14.3	12.6	101.6°	134.8°
Experiment	-4.2	9.0	102.4°	135.5°

The only method that describes the singlet-triplet energy splitting ΔE_{S-T} with almost chemical accuracy (1 kcal/mol) for both systems is CCSD(T). The other methods show

larger deviations from the experimental values. HF overestimates the singlet-triplet energy splitting for both systems and results in too large ΔE_{S-T} values. This is due to the fact that a triplet needs to be calculated using UHF, which contains spin contamination and therefore lowers the energy.

MP2 underestimates the ΔE_{S-T} for p-benzyne and overestimates the ΔE_{S-T} for methylene. Nevertheless, MP2 is the only method that describes the singlet-triplet energy splitting for p-benzyne with a negative value, which is in line with the experimental value and also gives a reasonable value for the ΔE_{S-T} of methylene. The large error in the p-benzyne ΔE_{S-T} is due to the small HOMO-LUMO gap, which leads to a large energy correction of the correlation energy (see Equation 1).

All the other methods show positive values for the singlet-triplet energy splitting ΔE_{S-T} of p-benzyne, which are not physically meaningful.

The H-C-H angles for methylene are also affected by the choice of method, but all methods predict a similar results for the angle and are in good agreement with the experimental values. It is to be noted that the angles for HF again have the largest deviations from the experimental values.

2 Basis Set Convergence

The convergence of different basis sets is an important aspect of quantum chemical calculations. In this report, the basis set convergence of the formic acid dimer is investigated. The structure of the formic acid monomer and dimer are shown in Figure 3 and were optimized using the TPSS-D3/def2-TZVP level of theory.

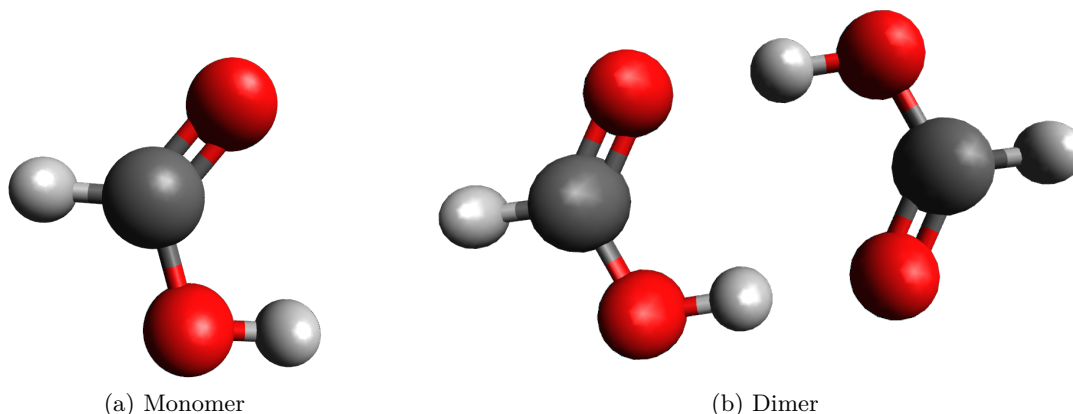


Figure 3: Monomer and dimer structures for formic acid. Geometries were optimized on the TPSS-D3/def2-TZVP level of theory.

The dimerization energy of formic acid is defined as the difference between the dimer energy and the sum of the monomer energies:

$$\Delta E_{\text{dimerization}} = E_{\text{dimer}} - 2 \cdot E_{\text{monomer}}, \quad (2)$$

where E_{dimer} is the energy of the dimer and E_{monomer} is the energy of the monomer.

The dimerization energy was computed for three different methods, namely HF, MP2 and TPSS-D3, using the cc-pVXZ^[6] and aug-cc-PVXZ basis sets, where X = D, T, Q. The dimerization energy was computed for each method and basis set combination with optimized geometries at the TPSS-D3/def2-TZVP level. The results are shown in Figure 4.

The dimerization energy for formic acid shows a strong dependence on the choice of basis set and method. The aug-cc-PVXZ basis sets generally yield more negative dimerization

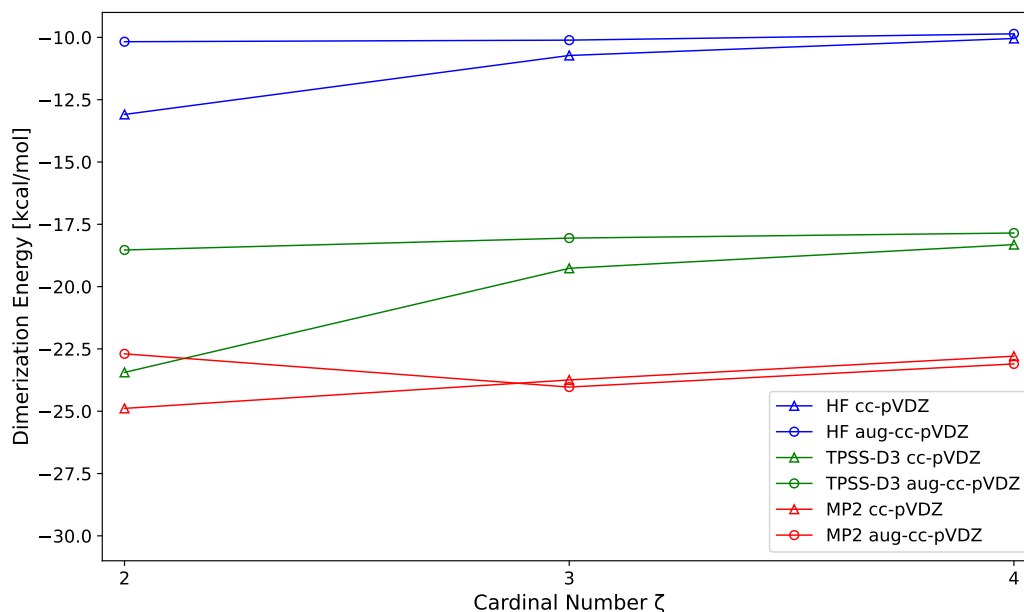


Figure 4: Convergence behaviour for the formic acid dimer aug-cc-PVXZ and cc-pVXZ basis sets at HF, MP2, and TPSS-D3 levels of theory.

energies compared to the cc-pVXZ basis sets. For a cardinal number of four, all methods and basis sets are at similar dimerization energies.

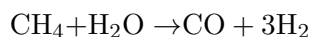
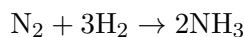
The calculations generally contains two different errors, the basis set incompleteness error (BSIE) and the basis set superposition error (BSSE). The BSIE is the error that arises from the finite size of the basis set and is the difference between the complete basis set (CBS) limit and the energy computed with a finite basis set. This error is always present and not only limited to dimerization energies. The BSIE can be reduced by using larger basis sets, but it is not possible to eliminate it completely. It is important to note that the convergence with respect to the basis set depends on the used method. For instance, SCF energies converge exponentially with respect to the basis set size, while correlation energies converge with inverse cubic behavior.

The BSSE is the error that arises from the interaction of the basis functions of the two monomers in the dimer. This error is present in energies for the dimer or other intramolecular interactions. The BSSE is generally larger for an increased basis sets, because the interaction between the basis functions of the two monomers is stronger. The BSSE compensates the BSIE and leads to a artificially lowered dimerization energy. The BSSE is only not present if one uses a minimal basis set, which is not the case here.

Both errors can be reduced by using larger basis sets, but it is not possible to eliminate them completely, only by theoretically using a CBS. Another possibility is the use of Boys-Bernardi counterpoise correction, which is a method to correct for the BSSE.

3 Thermochemistry

The reaction enthalpies ΔH for the Haber-Bosch process and the steam-reforming of methane were computed using the different level of theory.



The geometries were optimized on the TPSS-D3/def2-TZVP level of theory. In addition, the geometries of the molecules involved in the Haber Bosch process were also optimized at the MP2/def2-TZVP level of theory and the RMSD to the TPSS-D3/def2-TZVP geometries was computed. The results are shown in Table 2. The RMSD values are in the order of 10^{-3} Å and show that the geometries are very similar. This indicates that the choice of the method does not have a large impact on the geometry and the selection of the method can mainly be based on the computational cost.

Table 2: RMSD between TPSS-D3 and MP2 geometries optimized in a def2-TZVP basis for all molecules involved in the Haber-Bosch process.

	RMSD (Å)
N ₂	$5.67 \cdot 10^{-3}$
H ₂	$3.06 \cdot 10^{-3}$
NH ₃	$6.58 \cdot 10^{-3}$

The reaction enthalpies of both reactions were computed using the MP2 and B3LYP-D3 methods in a def2-TZVP basis set and The Becke 2-parameter exchange functional with Perdew-Lee-Yang-Parr correlation functional (B2PLYP)-D3 and CCSD(T) in a def2-QZVP basis set. First, the single point energies were computed using the optimized geometries from the TPSS-D3/def2-TZVP level of theory. The single point energies were then corrected for the thermal corrections at 298 K. The results are shown in Table 3.

In total, all methods yield a reasonable reaction enthalpies. The MP2 and the B3LYP-D3 methods underestimate the reaction enthalpies for both reactions, while the B2PLYP-D3 and CCSD(T) methods overestimate them.

The B2PLYP-D3 method yields the best agreement with the experimental values for both reactions, while the CCSD(T) method is slightly worse. The MP2 and B3LYP-D3

Table 3: Reaction enthalpies in kcal/mol for the Haber-Bosch process and steam-reforming of methane. Geometries were optimized on the TPSS-D3/def2-TZVP level. B2PLYP-D3 and CCSD(T) energies were computed in a def2-QZVP basis, def2-TZVP was used for all other methods.

Reaction	MP2	B3LYP-D3	B2PLYP-D3	CCSD(T)
Haber-Bosch	-12.1	-21.0	-23.9	-24.5
	Exp: ^[4]			-22.5
Steam-Reforming	43.1	47.3	50.4	51.4
	Exp: ^[4]			49.3

methods are not as accurate.

The general trend of the *Jacob's Ladder* is observed for this example, where the more accurate methods yield better results. The hybrid functional B3LYP-D3 is more accurate than the pure MP2 method, but not as accurate as the double hybrid B2PLYP-D3.

4 Kinetics

The kinetic isotope effect for the reaction of methane with a hydroxyl radical was studied using the B3LYP-D3^[7]/def2-TZVP level of theory for a transition state search, which was performed using TURBOMOLE.^[5]

The transition state research was done for $\text{CH}_4\text{-OH}$ and $\text{CD}_4\text{-OH}$, where the latter is the deuteriated isotopologue of methane. The geometry of the transition state is shown in Figure 5.

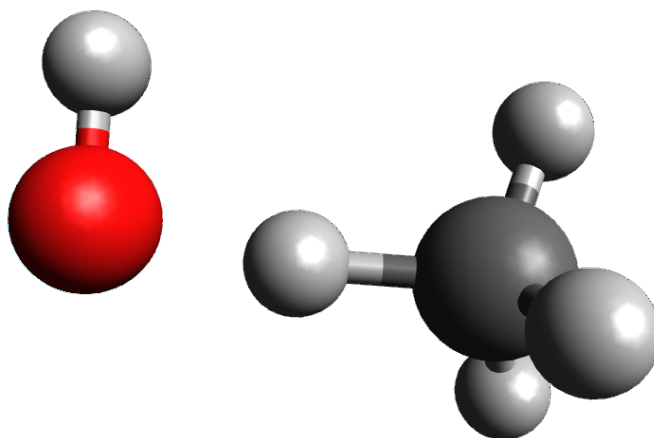


Figure 5: Geometry of the transition state for the reaction of methane with a hydroxy radical. The transition state was optimized on the B3LYP-D3/def2-TZVP level of theory.

The electronic energies, zero-point vibrational energy (ZPVE) and thermal corrections at 298 K for the reactants and transition states are shown in Table 4. The electronic energies were computed using the B3LYP-D3/def2-TZVP level of theory. The ZPVE and thermal corrections were computed using the same level of theory.

Table 4: Electronic energies (E_h) along with zero-point vibrational energies and thermal corrections (298 K) for reactants and transition states.

	CH_4	CD_4	OH	$\text{CH}_4\text{ - OH}$	$\text{CD}_4\text{ - OH}$
E_{el} [kcal/mol]	-25.415	-25.415	-47.524	-72.937	-72.937
ZPVE (kcal/mol)	27.959	20.500	5.256	31.749	25.217
Thermal correction (kcal/mol)	2.393	2.468	2.074	3.475	3.732

The energies in Table 4 amount to

$$\Delta H_H^\ddagger = 0.325 \text{ kcal/mol}$$

$$\Delta H_D^\ddagger = 1.434 \text{ kcal/mol}$$

with the subscripts H and D referring to regular and deuteriated methane respectively.

As seen in Table 4, the transition state for the reaction of methane with a hydroxy radical is lower in energy than the transition state for the reaction of deuteriated methane with a hydroxy radical due to the fact that the ZPVE and the thermal corrections are lower for the regular methane. The electronic energies are the same for both transition states, because they do not depend on the mass of the nuclei in the molecules, as long as relativistic effects are not neglected.

The thermal corrections in contrast depend on the mass of the nuclei, which is why the thermal correction for the transition state of deuteriated methane is larger than for regular methane. The vibrational frequencies, which are taken in account for the thermal correction, can be computed from the harmonic oscillator model:

$$\nu = \frac{1}{2\pi} \sqrt{\frac{k}{\mu}}, \quad (3)$$

where k is the force constant and μ is the reduced mass of the system. The reduced mass can be computed from the masses of the two atoms involved in the reaction:

$$\mu = \frac{m_1 \cdot m_2}{m_1 + m_2}. \quad (4)$$

The reduced mass for the transition state of regular methane is smaller than for the transition state of deuteriated methane, which leads to larger vibrational frequencies and therefore a smaller thermal correction.

The relative reaction rates can then be determined from $\Delta H^\ddagger$ according to:

$$\frac{k_H}{k_D} = \exp\left(\frac{\Delta H_D^\ddagger - \Delta H_H^\ddagger}{RT}\right) = 6.502 \quad (5)$$

This indicates that the reaction of methane with a hydroxy radical is about 6.5 times

faster than the reaction of deuteriated methane with a hydroxy radical.

5 Solvation

The S_N2 reaction of chloromethane with fluoride is a well-known reaction in organic chemistry. The reaction proceeds via a transition state, where the nucleophile attacks the electrophile and the leaving group is expelled.

The solvation effects on the reaction can be significant, as the presence of a solvent can stabilize certain states and alter the reaction pathway. In this case, we consider the solvation effects of methanol on the S_N2 reaction of chloromethane with fluoride. The geometry of the transition state in vacuum is shown in Figure 6.

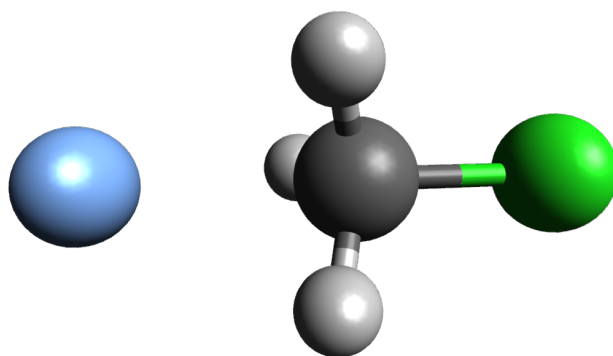


Figure 6: Geometry of the transition state for the S_N2 reaction of Chloromethane and Fluoride. The transition state was optimized on the PW6B95-D3/def2-TZVP level of theory. The C-F distance in the figure is 4.00 Å in vacuum.

The potential energy curves for the S_N2 reaction of chloromethane with fluoride in vacuum and in the presence of methanol as an implicit solvent were computed using the PW6B95^[8]-D3^[9]/def2-TZVP^[10] level of theory. Both calculations were performed using TURBOMOLE.^[5] The potential energy curves were computed for C-F distances between 2.25 Å and 8.0 Å in vacuum and in the presence of methanol as an implicit solvent using the COSMO model. The potential energy curves are shown in Figure 7.

The potential energy curves in Figure 7 show that the reaction proceeds via a transition state, where the C-F distance is about 3.75 Å in vacuum and about 4.00 Å in the presence of methanol as an implicit solvent. The transition state is lower in energy in the presence of methanol, which indicates that the solvation stabilizes the transition state and lowers the activation barrier for the reaction. The different C-F distances in vacuum and in the presence of methanol are due to the charge shielding effect of the solvent.

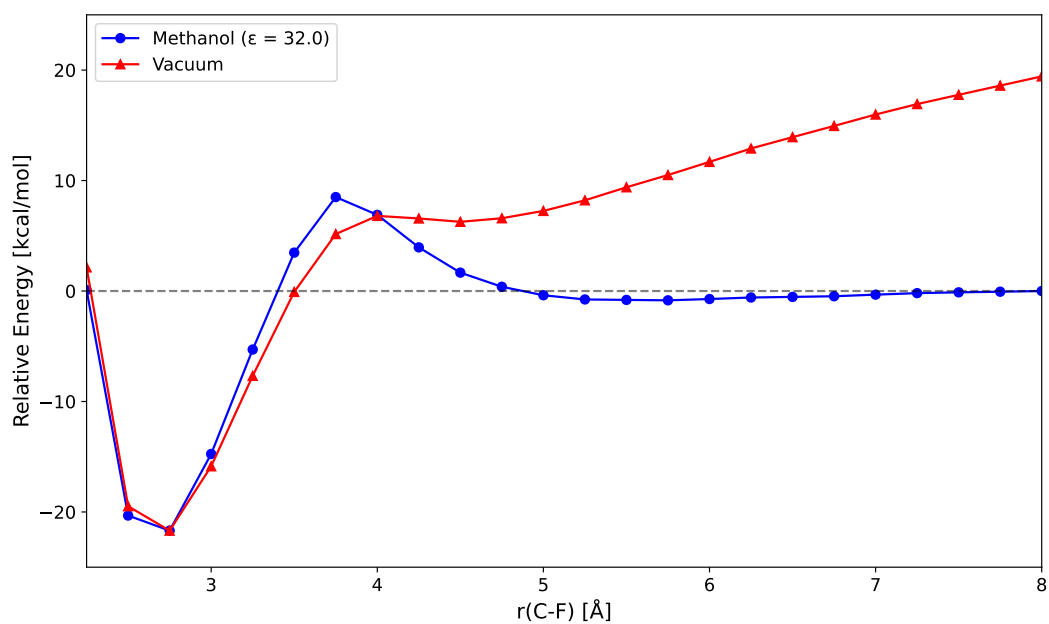


Figure 7: Relative energy curve for the S_N2 reaction of Chloromethane and Fluoride. Blue circles: In presence of methanol as an implicit solvent. Red triangles: In vacuum. The blue curve has been adjusted to the dissociation limit and the red curve was shifted such that both minima align.

The reaction barriers for the reactions can be estimated based on the potential curves:

$$\Delta E_{\text{solv}}^{\ddagger} = 9.3 \text{ kcal/mol}$$

$$\Delta E_{\text{gas}}^{\ddagger} = 0.5 \text{ kcal/mol}$$

The reaction barrier in the presence of methanol as an implicit solvent is significantly higher than in vacuum, which indicates that the solvation effects are significant and can alter the reaction pathway.

Nevertheless, the reaction barrier can not be compared this easy, because the fluoride ion in vacuum is not solvated, while the fluoride ion in the presence of methanol is solvated. The solvation of the fluoride ion in methanol leads to a stabilization of the fluoride ion, because the negative charge of the fluoride ion is shielded by the positive solvate molecules. In contrast, the fluoride ion in vacuum is not shielded and therefore unstable. As the distance between fluoride ion and the chloromethane molecule in vacuum decreases, the fluoride ion is stabilized by the chloromethane molecule. In the presence of methanol, the fluoride ion is already stabilized by the solvent and the energy curve converges towards the dissociation limit at a larger distance. This leads to a larger reaction barrier in the presence of methanol.

6 Activation Energies

The reaction path for the Claisen rearrangement of allyl-vinyl ether and the Diels-Alder reaction of cyclopentadiene to *endo*-dicyclopentadiene were computed using the double ended growing string method (GSM) at the GFN2-xTB^[11] level. First, geometries of the reactant and product were optimized using the GFN2-xTB method. Then, the double ended GSM was used to compute the reaction path between the reactant and product of both reactions for 20 points. The reaction paths are shown in Figure 8 and Figure 9.

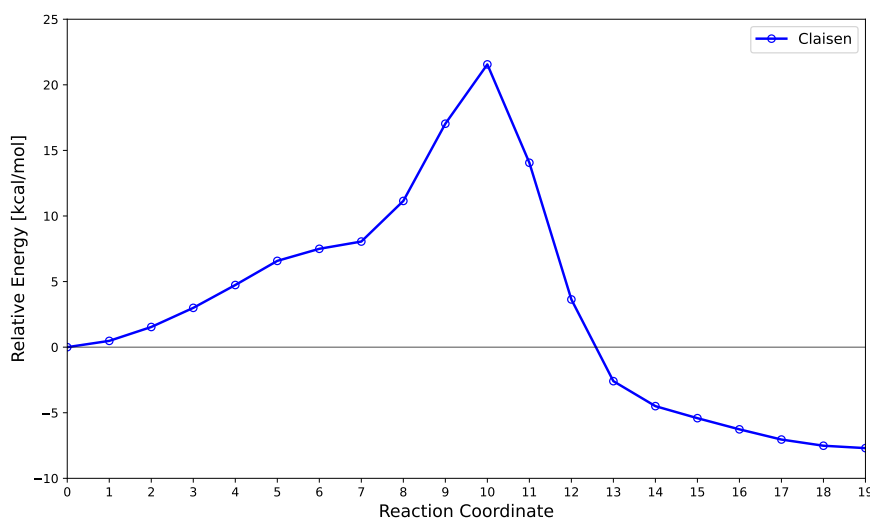


Figure 8: Relative energy along the reaction coordinate for the Claisen rearrangement of allyl-vinyl ether.

The activation energy for the Claisen rearrangement of allyl-vinyl ether is about 21.6 kcal/mol. The reaction proceeds via a cyclic transition state, where the C-O bond is broken and the C-C bond is formed. The transition state is lower in energy than the reactant and product, which indicates that the reaction is exothermic.

This activation energy is smaller than the experimental value of 30.6 kcal/mol,^[12] which suggests that the computational method used may underestimate the activation energy. This indicates that the transition state is not described accurately enough, which can be due to the limitations of the GFN2-xTB method. The GFN2-xTB method is a semi-empirical method that is not as accurate as ab initio methods, but it is computationally efficient and can be used for larger systems.

The activation energy for the Diels-Alder reaction of cyclopentadiene to *endo*-dicyclopentadiene

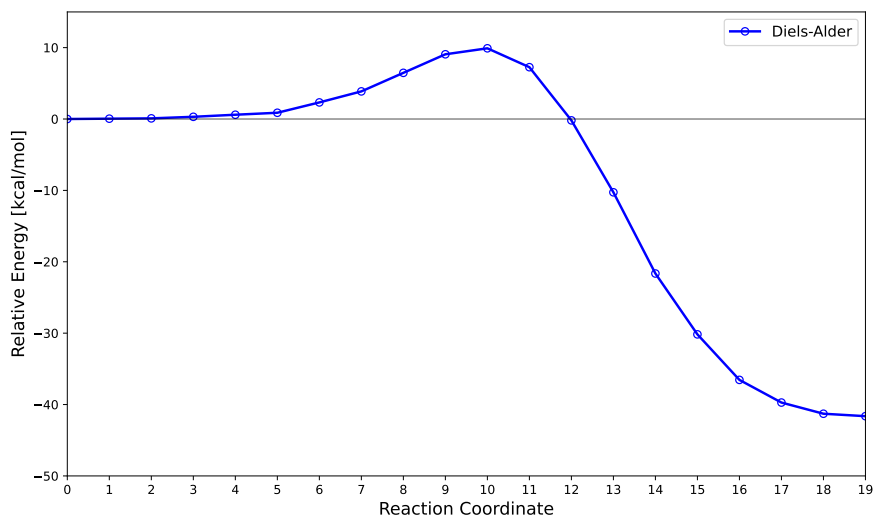


Figure 9: Relative energy along the reaction coordinate for the Diels-Alder reaction of cyclopentadiene to *endo*-dicyclopentadiene.

is about 9.9 kcal/mol. The reaction proceeds via a transition state, where the C-C bond is formed and the C=C bond is broken. The transition state is lower in energy than the reactant and product, which indicates that the reaction is exothermic.

The reliability of the activation energy for both reactions can be improved by using a higher level density functional theory (DFT) method. This would provide a more accurate description of the reaction path, but would also increase the computational cost. Further improvements can be made by optimizing the input geometries of the reactants and products, as this has a large impact on the used method.

7 Noncovalent Interactions

Noncovalent interactions are important in many chemical and biological processes. In this section, the interaction of methane with argon and krypton is investigated. The geometry of the methane-argon complex is shown in Figure 10.

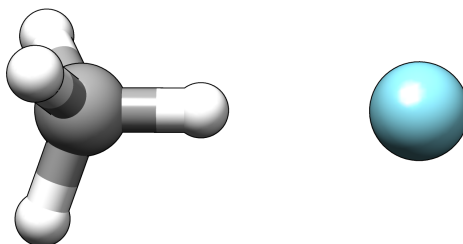


Figure 10: Geometry of the $\text{CH}_4 \cdots \text{Ar}$ complex. Figure taken from [4].

The potential energy curves for the interaction of methane with argon and krypton were computed using the following methods: Becke-Lee-Yang-Parr (BLYP), BLYP-D3 and MP2. All calculations were performed using the def2-QZVP basis set. The potential energy curves were computed for $\text{CH}_4\text{-Ar/Kr}$ distances between 4.5 Å and 15.0 Å, were normalized to the dissociation limit and are shown in Figure 11.

The potential energy curves in Figure 11 show the importance of dispersion interactions for noncovalent interactions. The BLYP method does not account for dispersion interactions and therefore yields a repulsive potential energy curve, which is not physically meaningful.

The BLYP-D3 method accounts for dispersion interactions and yields a potential energy curve that is in good agreement with the MP2 method. The MP2 method yields a potential energy curve that is slightly lower in energy than the BLYP-D3 method. Both methods yield a minimum at around 6.0 Å for the $\text{CH}_4\text{-Ar}$ complex and for the $\text{CH}_4\text{-Kr}$ complex.

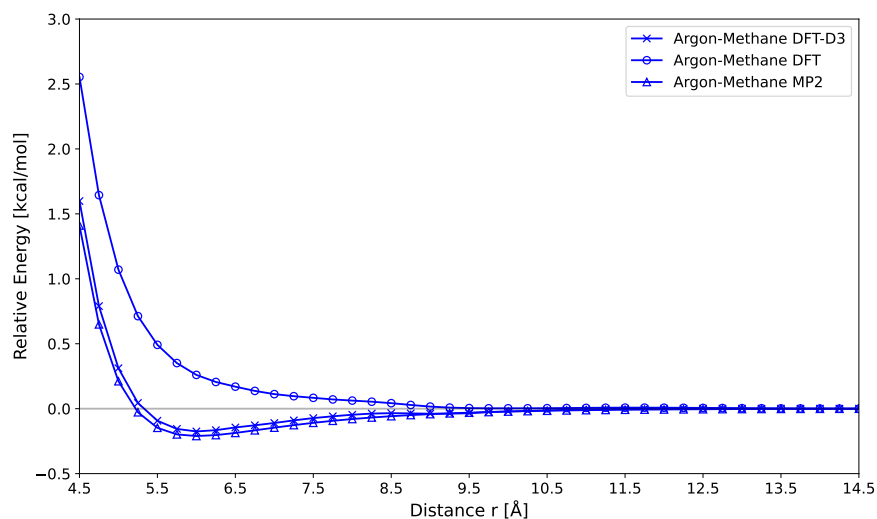
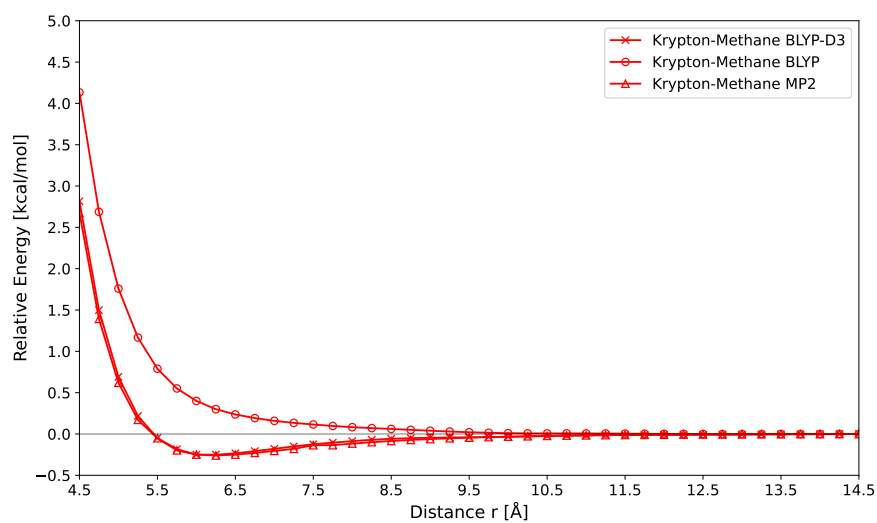
(a) $\text{CH}_4 \cdots \text{Ar}$ (b) $\text{CH}_4 \cdots \text{Kr}$

Figure 11: Potential energy curves for methane with argon (a) and krypton (b).

8 Spectroscopy

8.1 IR-Spectroscopy

The IR-spectra of 1,4-Benzoquinone were computed on the TPSS-D3/def2-SVP and HF-D3/def2-SVP levels of theory using TURBOMOLE.^[5] The structure of 1,4-Benzoquinone is shown in Figure 12.

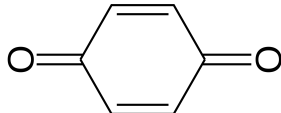


Figure 12: Structure of 1,4-Benzoquinone.

First, the geometry of 1,4-Benzoquinone was optimized with TURBOMOLE^[5] using the respective level of theory. Then, the vibrational modes were computed using the subprogram aoforce. The resulting IR-modes are shown in Table 5. Furthermore, the scaling factor for the IR-modes was calculated using the experimental values from [4] and the theoretical values from the TPSS-D3/def2-SVP and HF-D3/def2-SVP levels of theory. The scaling factor was calculated using the following equation:

$$f_{\text{scal}} = \frac{\tilde{\nu}_{\text{exp}}}{\tilde{\nu}_{\text{theo}}}, \quad (6)$$

where $\tilde{\nu}_{\text{exp}}$ is the experimental wavenumber and $\tilde{\nu}_{\text{theo}}$ is the theoretical wavenumber.

Table 5: Wavenumbers of IR-modes (cm^{-1}) calculated at the HF-D3/def2-SVP and TPSS-D3/def2-SVP level along an experimental reference [4]. Each theoretical frequency has been assigned a scaling factor (see Eq. (7))

Exp ^[4]		HF-D3			TPSS-D3		
$\tilde{\nu}$	I	$\tilde{\nu}$	f_{scal}	I_{rel}	$\tilde{\nu}$	f_{scal}	I_{rel}
3075.0	0.01	3387.6	0.91	0.01	3133.74	0.98	0.01
1670.0	1	2028.65	0.82	1	1704.54	0.98	1
1290.0	0.23	1452.21	0.89	0.17	1304.92	0.99	0.29
1070.0	0.15	1167.24	0.92	0.06	1036.34	1.03	0.15
870	0.3	988.47	0.88	0.17	885.49	0.98	0.23

The averaged scaling factors for the IR-modes were found to be 0.88 for HF-D3 and 0.99 for TPSS-D3. This indicates that the TPSS-D3 method yields more accurate results than

the HF-D3 method for the IR-spectra of 1,4-Benzoquinone. Both methods overestimate the wavenumbers, but the TPSS-D3 method is almost identical to the experimental values. This large overestimation for HF-D3 is due to the underestimation of the bond strength. TPSS-D3 is a meta-GGA that includes the kinetic energy of the electrons inside the exchange correlation potential, which leads to a better description of the bond strength and therefore a better description of the IR-spectra.

The calculations could be improved by using a larger basis set or using a higher level of theory. However, this would increase the computational cost significantly.

8.2 The Color of Indigo

The color of indigo is a well-known example of a molecule that exhibits strong color due to its electronic structure. The structure of indigo was optimized using the TPSS-D3/def2-SVP level of theory and is shown in Figure 13.

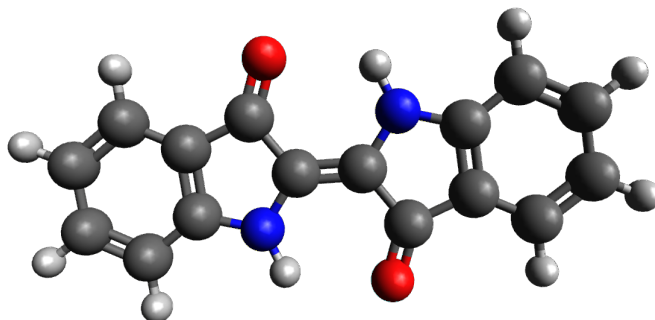


Figure 13: Optimized geometry of indigo on the TPSS-D3/def2-SVP level of theory.

The optical transitions of indigo were calculated using time-dependent HF method and time-dependent DFT methods with the Perdew-Burke-Ernzerhof (PBE) and Perdew-Burke-Ernzerhof with 25% Hartree-Fock exchange (PBE0) functionals in a def2-SVP basis set. The results are shown in Table 6.

Table 6: Optical transition the indigo molecule calculated using different methods in the def2-SVP basis set.

	HF	PBE	PBE0	Exp
λ (nm)	373.9	618.7	533.5	610
Colour	yellow/orange	blue	violet	blue

The PBE and PBE0 methods yield a blue and violet color, respectively, while HF yields a yellow/orange color. The experimental value is in good agreement with the PBE method.

The PBE0 method is not in agreement with the experimental value. The HF method also does not match the experimental value and has even more significant deviations than PBE0. This indicates that the HF method is not suitable for the calculation of optical transitions of indigo at all.

The PBE method is a GGA functional that should perform worse than the hybrid PBE0 functional according to *Jacob's Ladder*, but in this case it yields a better result.

8.3 NMR Parameters

^{13}C NMR Shifts for Organic Compounds

The ^{13}C NMR shifts for several organic compounds were calculated using the PBEh-3c method for geometry optimization and the PBE/pcSeg-2 method for the calculation of the shielding tensors in ORCA 5.0.4.^[1,2] The Lewis structures of the compounds are shown in Figure 14.

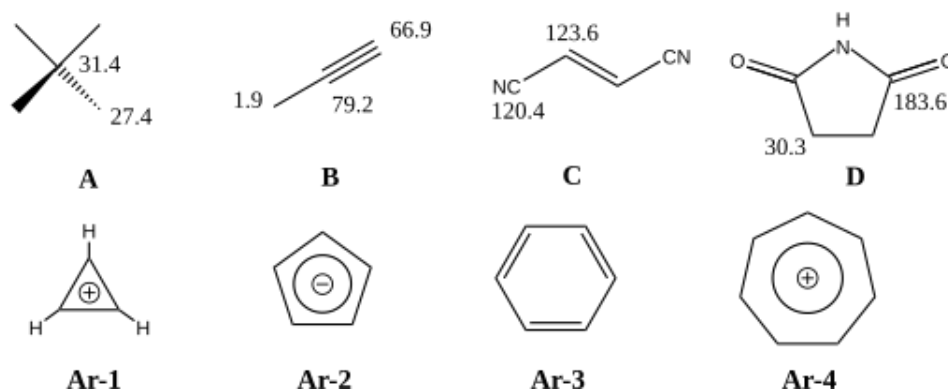


Figure 14: Lewis structures of four organic compounds A – D with their experimentally obtained chemical shift as well as four aromatic compounds Ar-1 - Ar-4. Figure taken from [4].

The chemical shifts were calculated relative to tetramethylsilane (TMS) and the results are shown in Table 7.

The mean deviation (MD) and mean absolute deviation (MAD) values for the chemical shifts of the organic compounds are:

$$\text{MD} = \frac{1}{N} \sum_{i=1}^N (\delta_{\text{calc}} - \delta_{\text{exp}}) = 4.00 \text{ ppm}$$

$$\text{MAD} = \frac{1}{N} \sum_{i=1}^N |\delta_{\text{calc}} - \delta_{\text{exp}}| = 4.33 \text{ ppm.}$$

The similarity of the MD and MAD values indicates that the method has a general difference in the chemical shifts, but the deviations are not too large. The largest

Table 7: ^{13}C chemical shifts (δ) for several organic molecules in reference to TMS. Geometries were optimized using PBEh-3c and shielding tensors were calculated at the PBE/pcSeg-2 level of theory.

Compound		δ (ppm)		
A	Calc	34.5	34.0	
	Exp	27.4	31.4	
B	Calc	4.03	71.2	85.4
	Exp	1.9	66.9	79.2
C	Calc	124.0	127.9	
	Exp	120.4	123.6	
D	Calc	33.9	182.3	
	Exp	30.3	183.6	

deviation is for compound A, which has a difference of 7.1 ppm, and the smallest deviation is for compound D, which has a difference of -1.3 ppm.

The aromatic compounds Ar-1 - Ar-4 were also calculated using the PBE/pcSeg-2 method and their chemical shifts are shown in relation to their π -electron density ρ_π in Figure 15. The chemical shifts were calculated relative to TMS.

The chemical shifts of the aromatic compounds show a linear correlation with the π -electron density, which is expected due to the fact that the chemical shift is influenced by the electron density around the nucleus. The magnetic moment created by the aromatic ring current leads to an increased shielding of the nucleus, which results in a downfield shift of the chemical shift.

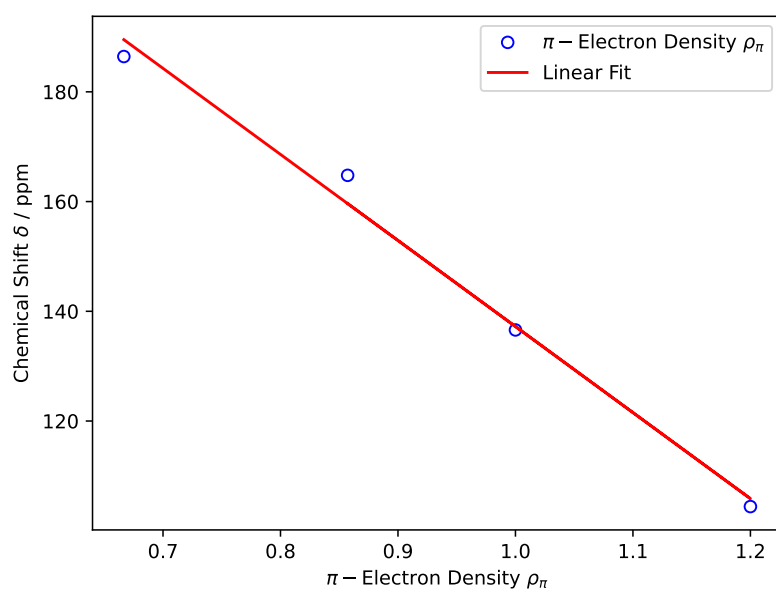


Figure 15: Chemical shifts δ of the aromatic compounds Ar-1 - Ar-4 plotted against the π -Electron Density ρ_π . The linear fit is shown as red line and the calculated values as blue circles. The chemical shifts were calculated using the PBE/pcSseg-2 level of theory.

^{13}C and ^{17}O NMR Shifts for Acetone

The effect of the solvent on the NMR shifts of acetone was investigated by calculating the chemical shieldings for acetone in gas-phase and in a solvent (water) using the PBE/pcSeg-2 method. The geometries were optimized in gas-phase and in a solvent using the COSMO model. The chemical shieldings were calculated relative to TMS for ^{13}C and relative to water for ^{17}O . The results are shown in Table 8.

Table 8: Chemical shieldings for Acetone relative to TMS (^{13}C) and water (^{17}O) (δ in ppm). Reference data was optimized and calculated in gas-phase.

Geometry	NMR	^{13}C		^{17}O	
		δ	$\Delta\delta_{\text{gas/gas}}$	δ	$\Delta\delta_{\text{gas/gas}}$
Gas	Gas	214.01	0.00	672.54	0.00
Gas	Solvent	224.08	10.07	616.54	-60.62
Solvent	Gas	216.17	2.16	682.76	10.23
Solvent	Solvent	226.25	12.24	618.83	-53.71

The difference in the chemical shifts for the different combinations of gas-phase and solvent-phase geometries is 2.16 ppm for ^{13}C and 10.23 ppm for ^{17}O . In comparison to the absolute value, this gives differences of 1.0-1.5%. This indicates that the solvent has a significant effect on the chemical shifts of acetone, especially for ^{17}O . The chemical shift for ^{13}C is less affected by the solvent, but still shows a significant difference.

The differences in the chemical shifts for gas-phase and solvent-phase for the NMR calculations are larger than that for the geometry. This indicates that the choice for the NMR calculation is more significant than the choice for the geometry.

^1H NMR Shifts for Metacyclophane

The ^1H NMR shifts for in- and out-Metacyclophane were calculated using the PBEh-3c method for geometry optimization and the PBE/pcSeg-2 method for the calculation of the shielding tensors in ORCA 5.0.4.^[1,2] The optimized geometries on the PBE-3c method of in- and out-Metacyclophane are shown in Figure 16.

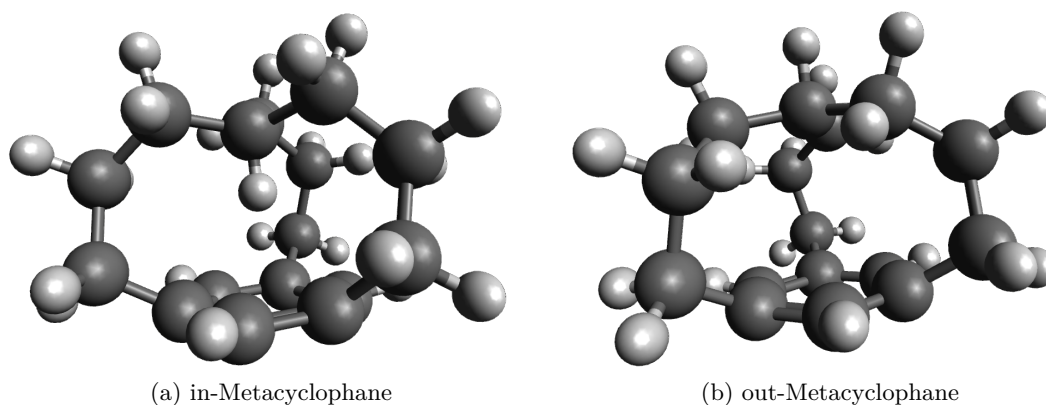


Figure 16: Optimized geometry of in- and out- $[3^{4,10}][7]$ Metacyclophane on the PBE-3c level.

The chemical shifts were calculated relative to TMS and are:

$$\text{in: } \delta(^1\text{H}): -3.24$$

$$\text{out: } \delta(^1\text{H}): 2.07.$$

The chemical shifts for in-Metacyclophane are negative, which indicates that the protons are shielded by the aromatic ring current. The chemical shifts for out-Metacyclophane are positive, which indicates that the protons are deshielded by the aromatic ring current.

9 Appendix

Acronyms

B2PLYP Becke 2-parameter exchange functional with Perdew-Lee-Yang-Parr correlation functional

B3LYP Becke 3-parameter exchange functional with Lee-Yang-Parr correlation functional

BLYP Becke-Lee-Yang-Parr

BSIE basis set incompleteness error

BSSE basis set superposition error

CASSCF complete active space self-consistent field

CBS complete basis set

CCSD(T) coupled cluster with single, double and perturbative triple excitations

DFT density functional theory

GSM growing string method

HF Hartree-Fock

MAD mean absolute deviation

MD mean deviation

MP2 Møller-Plesset perturbation theory of second order

PBE Perdew-Burke-Ernzerhof

PBE0 Perdew-Burke-Ernzerhof with 25% Hartree-Fock exchange

PW6B95 Perdew-Wang 6-parameter exchange functional with Becke-95 correlation

RHF restricted Hartree-Fock

TMS tetramethylsilane

TPSS Tao-Perdew-Staroverov-Scuseria

UHF unrestricted Hartree-Fock

ZPVE zero-point vibrational energy

References

- [1] Frank Neese. “The ORCA Program System.” In: *WIREs Computational Molecular Science* 2.1 (2012), pp. 73–78.
- [2] Frank Neese. “Software Update: The ORCA Program System—Version 5.0.” In: *WIREs Computational Molecular Science* 12.5 (2022), p. 1606.
- [3] Björn O. Roos, Peter R. Taylor, and Per E. M. Sigbahn. “A Complete Active Space SCF Method (CASSCF) Using a Density Matrix Formulated Super-CI Approach.” In: *Chemical Physics* 48.2 (), pp. 157–173.
- [4] AK Grimme. *Quantum Chemistry II Script*. Quantum Chemistry II script. 2025. URL: <https://qc2-teaching.readthedocs.io/en/latest/apps-ex.html#>.
- [5] University of Karlsruhe and Forschungszentrum Karlsruhe GmbH. *TURBOMOLE V7.6*. TURBOMOLE GmbH, 2007.
- [6] Thom H. Dunning Jr. “Gaussian Basis Sets for Use in Correlated Molecular Calculations. I. The Atoms Boron through Neon and Hydrogen.” In: *The Journal of Chemical Physics* 90.2 (1989), pp. 1007–1023.
- [7] Axel D. Becke. “Density-functional Thermochemistry. III. The Role of Exact Exchange.” In: *The Journal of Chemical Physics* 98.7 (1993), pp. 5648–5652.
- [8] Yan Zhao and Donald G. Truhlar. “Design of Density Functionals That Are Broadly Accurate for Thermochemistry, Thermochemical Kinetics, and Nonbonded Interactions.” In: *The Journal of Physical Chemistry A* 109.25 (2006), pp. 5656–5667.
- [9] Stefan Grimme, Stephan Ehrlich, and Lars Goerigk. “Effect of the Damping Function in Dispersion Corrected Density Functional Theory.” In: *Journal of Computational Chemistry* 32.7 (2011), pp. 1456–1465.

- [10] Florian Weigend and Reinhart Ahlrichs. “Balanced Basis Sets of Split Valence, Triple Zeta Valence and Quadruple Zeta Valence Quality for H to Rn: Design and Assessment of Accuracy.” In: *Physical Chemistry Chemical Physics* 7.18 (2023), pp. 3297–3305.
- [11] Christoph Bannwarth, Sebastian Ehlert, and Stefan Grimme. “GFN2-xTB—An Accurate and Broadly Parametrized Self-Consistent Tight-Binding Quantum Chemical Method with Multipole Electrostatics and Density-Dependent Dispersion Contributions.” In: *Journal of Chemical Theory and Computation* 15.3 (2019), pp. 1652–1671.
- [12] Frederic W. Schuler and George W. Murphy. “The Kinetics of the Rearrangement of Vinyl Allyl Ether.” In: *Journal of the American Chemical Society* 72.7 (1950), pp. 3155–3159.